

Solved selected problems of General Relativity - Thomas A. Moore

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Chapter 20 - The Stress-Energy Tensor

Solution. **BOX 20.1** - Exercise 20.1.1.

If the source of the gravitational field of the box were the total energy, then we would not be firing particles to infinity with energy we create out of nothing, since after the matter-antimatter combination, the photons created would hold kinetic energy, which would still be in the box because it's perfectly mirrored, in short, because of the conservation of energy, we cannot create energy out of nothing. $\square$

Solution. **BOX 20.2** - Exercise 20.2.1.

Let us consider the Stress-Energy components $T^{ij} = \rho_0 u^i u^j$ where i and j are both spatial indices, then we see that

$$\begin{aligned} T^{ij} &= \rho_0 u^i u^j = n_0 m u^i u^j = n_0 m (u^t v_i) u^j = (n_0 u^t) v_i (m u^j) = \\ &= n v_i p^j = \frac{(n A v_i dt) p^j}{A dt} \end{aligned}$$

Therefore T^{ij} is the i-flux of j-momentum. $\square$

Solution. **BOX 20.3** - Exercise 20.3.1.

If we consider the component T^{xx} then the integral over p^x becomes

$$\int_{-\infty}^{\infty} f([p^x]^2)(p^x)^2 dp^x$$

We note that the integrand is an even function, so the integral along a symmetric interval is not necessarily 0, as opposed to the case where the integrand is an odd function like $f([p^x]^2)p^x$. $\square$

Solution. **BOX 20.3** - Exercise 20.3.2.

Since the selection of the x direction was arbitrary must be true for any other direction. Hence by symmetry we have that $T^{xx} = T^{yy} = T^{zz}$. $\square$

Solution. **BOX 20.3** - Exercise 20.3.3.

We note that $N(|\vec{p}|)/L^3$ is the number density of particles having momentum $\vec{p}$ and p^t is the energy of a single particle with momentum $\vec{p}$, then $N(|\vec{p}|)p^t/L^3$ is the energy density of particles with momentum $\vec{p}$.

Then, in the integral

$$T^{tt} = \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) p^t$$

We are summing across the whole momentum space i.e. we are considering all possible momentums inside the box so we get that $T^{tt} = \rho$ where ρ is the total energy density inside the box. $\square$

Solution. **BOX 20.4** - Exercise 20.4.1.

In a LIF where the fluid at a given event is (at least instantaneously) at rest the metric $g_{\mu\nu} = \eta_{\mu\nu}$ and the fluid's four-velocity at that event is $u^\mu = [1, 0, 0, 0]$.

Given that $T^{\mu\nu} = (\rho_0 + p_0)u^\mu u^\nu + p_0 g^{\mu\nu}$ then if $\mu \neq \nu$ then $T^{\mu\nu} = 0$ because the first term vanishes and the only non-zero components of $\eta^{\mu\nu}$ are the ones where $\mu = \nu$.

If $\mu = \nu = t$ we see that

$$T^{tt} = (\rho_0 + p_0)u^t u^t + p_0 \eta^{tt} = (\rho_0 + p_0) \cdot 1 + p_0 \cdot (-1) = \rho_0$$

And if $\mu = \nu = x$ or y or z we get that

$$T^{xx} = (\rho_0 + p_0)u^x u^x + p_0 \eta^{xx} = 0 + p_0 \cdot 1 = p_0$$

$$T^{yy} = (\rho_0 + p_0)u^y u^y + p_0 \eta^{yy} = 0 + p_0 \cdot 1 = p_0$$

$$T^{zz} = (\rho_0 + p_0)u^z u^z + p_0 \eta^{zz} = 0 + p_0 \cdot 1 = p_0$$

Therefore

$$T^{\mu\nu} = \begin{bmatrix} \rho_0 & 0 & 0 & 0 \\ 0 & p_0 & 0 & 0 \\ 0 & 0 & p_0 & 0 \\ 0 & 0 & 0 & p_0 \end{bmatrix}$$

□

Solution. **BOX 20.5** - Exercise 20.5.1.

Equation (20.29) states that

$$u^\mu u^\nu \partial_\mu (\rho_0 + p_0) + (\rho_0 + p_0) [u^\nu \partial_\mu u^\mu + u^\mu \partial_\mu u^\nu] + \eta^{\mu\nu} \partial_\mu p_0 = 0$$

So multiplying the equation by u_ν we get that

$$\begin{aligned} u_\nu u^\mu u^\nu \partial_\mu (\rho_0 + p_0) + (\rho_0 + p_0) [u_\nu u^\nu \partial_\mu u^\mu + u_\nu u^\mu \partial_\mu u^\nu] + u_\nu \eta^{\mu\nu} \partial_\mu p_0 &= 0 \\ -u^\mu \partial_\mu (\rho_0 + p_0) + (\rho_0 + p_0) [-\partial_\mu u^\mu + u_\nu u^\mu \partial_\mu u^\nu] + u^\mu \partial_\mu p_0 &= 0 \end{aligned}$$

Where we used that $u^\nu u_\nu = -1$.

Now, using that $u_\nu \partial_\mu u^\nu = 0$ we see that

$$\begin{aligned} -u^\mu \partial_\mu (\rho_0 + p_0) + (\rho_0 + p_0) [-\partial_\mu u^\mu + 0] + u^\mu \partial_\mu p_0 &= 0 \\ -u^\mu \partial_\mu \rho_0 - u^\mu \partial_\mu p_0 - \rho_0 \partial_\mu u^\mu - p_0 \partial_\mu u^\mu + u^\mu \partial_\mu p_0 &= 0 \\ -u^\mu \partial_\mu \rho_0 - \rho_0 \partial_\mu u^\mu - p_0 \partial_\mu u^\mu &= 0 \\ \partial_\mu (\rho_0 u^\mu) + p_0 \partial_\mu u^\mu &= 0 \end{aligned}$$

□

Solution. **BOX 20.5** - Exercise 20.5.2.

Multiplying the equation $\partial_\mu (\rho_0 u^\mu) + p_0 \partial_\mu u^\mu = 0$ by u^ν we get that

$$u^\nu \partial_\mu (\rho_0 u^\mu) + u^\nu p_0 \partial_\mu u^\mu = 0$$

Subtracting this equation from equation (20.29) we get that

$$\begin{aligned} u^\mu u^\nu \partial_\mu (\rho_0 + p_0) + (\rho_0 + p_0) [u^\nu \partial_\mu u^\mu + u^\mu \partial_\mu u^\nu] + \eta^{\mu\nu} \partial_\mu p_0 \\ - u^\nu u^\mu \partial_\mu \rho_0 - u^\nu \rho_0 \partial_\mu u^\mu - u^\nu p_0 \partial_\mu u^\mu &= 0 \\ u^\mu u^\nu \partial_\mu p_0 + (\rho_0 + p_0) u^\mu \partial_\mu u^\nu + \eta^{\mu\nu} \partial_\mu p_0 &= 0 \\ (u^\mu u^\nu + \eta^{\mu\nu}) \partial_\mu p_0 + (\rho_0 + p_0) u^\mu \partial_\mu u^\nu &= 0 \end{aligned}$$

Therefore

$$(\rho_0 + p_0) u^\mu \partial_\mu u^\nu = -(u^\mu u^\nu + \eta^{\mu\nu}) \partial_\mu p_0$$

□

Solution. P20.1

Let us consider the ionized hydrogen plasma at the sun's center, where the density is 160 g/cm^3 and the temperature is $1.5 \times 10^7 \text{ K}$.

We can compute the energy of an individual electron in the plasma by

$$\begin{aligned} p_e^t &= m_e + \frac{3}{2} k_B T \\ &= 9.109 \times 10^{-31} \text{ kg} + \frac{3}{2} \cdot 1.536 \times 10^{-49} \text{ kg/K} \cdot 1.5 \times 10^7 \text{ K} \\ &\approx 9.109 \times 10^{-31} \text{ kg} \end{aligned}$$

In the same way for a proton we get that $p_p^t \approx m_p = 1.672 \times 10^{-27} \text{ kg}$. Also, we can compute the number density of particles as follows

$$n = 160 \frac{\text{g}}{\text{cm}^3} \frac{2 \cdot 6.022 \times 10^{23} \text{ particles}}{1 \text{ g}} = 1.927 \times 10^{26} \frac{\text{particles}}{\text{cm}^3}$$

Since the particles are not moving relativistically then we may assume $u^t = 1$ and hence $\rho = np^t = \rho_0$ so the energy density is

$$\begin{aligned} \rho_0 &= \frac{n}{2} p_e^t + \frac{n}{2} p_p^t \\ &= \frac{1.927 \times 10^{26}}{2} \cdot 9.109 \times 10^{-31} + \frac{1.927 \times 10^{26}}{2} \cdot 1.672 \times 10^{-27} \\ &= 8.776 \times 10^{-5} + 0.161 \\ &= 0.16108 \text{ kg/cm}^3 = 161.08 \text{ g/cm}^3 \end{aligned}$$

We note that $\rho_0 \approx \text{mass density}$.

Now, using the ideal gas law we see that

$$\begin{aligned} p_0 &= \frac{N}{V} k_B T \\ &= nk_B T \\ &= 1.927 \times 10^{26} \cdot 1.536 \times 10^{-46} \cdot 1.5 \times 10^7 \\ &= 4.439 \times 10^{-13} \text{ g/cm}^3 \end{aligned}$$

Therefore we note that $p_0 \ll \rho_0$ and we have all the components of the stress-energy tensor. $\square$

Solution. **P20.2**

From equation (20.21) we know that $p_0 = T^{xx} = \rho_0(u^x)^2$ we want that $\rho_0 = 100p_0$ so we get that

$$p_0 = 100p_0(u^x)^2$$
$$u^x = \sqrt{\frac{1}{100}} = 0.1$$

In the same way, we see that $u^y = u^z = 0.1$. Then we get that

$$v_x = u^x \sqrt{1 - v^2} = 0.1 \sqrt{1 - v^2}$$

In the same way we have that $v_y = v_z = 0.1 \sqrt{1 - v^2}$. Then we solve for v as follows

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$$
$$v = \sqrt{3 \cdot 0.01(1 - v^2)}$$
$$v^2 = 0.03 - 0.03v^2$$
$$v^2(1 + 0.03) = 0.03$$
$$v = \sqrt{\frac{0.03}{1.03}} = 0.171$$

□

Solution. **P20.3**

Given that the star's pressure is negligible then equation (20.16) becomes

$$T^{\mu\nu} = \rho u^\mu u^\nu$$

Also, the velocity at any point of the star, in spherical coordinates has the form

$$\mathbf{v} = r\omega \sin \theta \hat{\boldsymbol{\varphi}}$$

or in cartesian coordinates (the LIF coordinates)

$$\begin{aligned} \mathbf{v} &= r\omega \sin \theta (-\sin \varphi \hat{\mathbf{x}} + \cos \varphi \hat{\mathbf{y}}) \\ &= -y\omega \hat{\mathbf{x}} + x\omega \hat{\mathbf{y}} \end{aligned}$$

Where we used that $x = r \sin \theta \cos \varphi$ and $y = r \sin \theta \sin \varphi$.

Then, the four-velocity u^α of the star, ignoring the terms of order v^2 , gives us

$$u^\alpha = \begin{bmatrix} 1 \\ v_x \\ v_y \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -y\omega \\ x\omega \\ 0 \end{bmatrix}$$

Therefore the stress-energy tensor for the rotating star is

$$T^{\mu\nu} = \begin{bmatrix} \rho & -\rho y\omega & \rho x\omega & 0 \\ -\rho y\omega & \rho y^2 \omega^2 & -\rho yx\omega^2 & 0 \\ \rho x\omega & -\rho yx\omega^2 & \rho x^2 \omega^2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

But this form of the stress-energy tensor still has components of order v_x^2 and v_y^2 , so dropping them we get that

$$T^{\mu\nu} = \begin{bmatrix} \rho & -\rho y\omega & \rho x\omega & 0 \\ -\rho y\omega & 0 & 0 & 0 \\ \rho x\omega & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

□

Solution. P20.4

Let us consider an ideal gas of photons in a perfectly-mirrored box in a LIF at rest with respect to the box.

We know that

$$T^{\mu\nu} = \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) \frac{p^\mu p^\nu}{p^t}$$

In particular if $\mu = \nu = t$ then we have that

$$T^{tt} = \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) p^t$$

We know that $N(|\vec{p}|)/L^3$ is the number density of photons having momentum $\vec{p}$ and $p^t = E$ is the energy of a single photon with momentum $\vec{p}$, then $N(|\vec{p}|)p^t/L^3$ is the energy density of particles with momentum $\vec{p}$.

Then the integral is the sum of all energy densities across the whole momentum space so $T^{tt} = \rho$ where ρ is the total energy density inside the box. Now, if $\mu = \nu = x$ we have that

$$\begin{aligned} T^{xx} &= \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) \frac{p^x p^x}{p^t} \\ &= \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) \frac{E^2 v_x^2}{E} \\ &= \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) E v_x^2 \end{aligned}$$

Where we used that $p^x = E v_x$, in the same way we see that

$$\begin{aligned} T^{yy} &= \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) E v_y^2 \\ T^{zz} &= \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) E v_z^2 \end{aligned}$$

Hence

$$\begin{aligned} T^{xx} + T^{yy} + T^{zz} &= \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) E (v_x^2 + v_y^2 + v_z^2) \\ &= \frac{1}{L^3} \iiint dp^x dp^y dp^z N(|\vec{p}|) E \\ &= \rho \end{aligned}$$

Where we used that $v_x^2 + v_y^2 + v_z^2 = 1$ for photons.

Also, we know that for an ideal gas $T^{xx} = T^{yy} = T^{zz}$, so we get that

$$T^{xx} = \frac{1}{3}\rho \quad T^{yy} = \frac{1}{3}\rho \quad T^{zz} = \frac{1}{3}\rho$$

We see that the same mathematical argument works to show that all off-diagonal elements in the stress-energy tensor are 0.

Therefore the stress-energy tensor for an ideal gas of photons is

$$T^{\mu\nu} = \begin{bmatrix} \rho & 0 & 0 & 0 \\ 0 & \frac{1}{3}\rho & 0 & 0 \\ 0 & 0 & \frac{1}{3}\rho & 0 \\ 0 & 0 & 0 & \frac{1}{3}\rho \end{bmatrix}$$

□

a. We know that

$$F^{\mu\nu} = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{bmatrix}$$

Before computing $F^{\mu\nu}F_{\mu\nu}$ we need to note first that in a LIF $F_{\mu\nu}$ is given by

$$F_{\alpha\beta} = \eta_{\alpha\mu}\eta_{\beta\nu}F^{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & B_z & -B_y \\ E_y & -B_z & 0 & B_x \\ E_z & B_y & -B_x & 0 \end{bmatrix}$$

Now, we see that

$$\begin{aligned} F^{\mu\nu}F_{\mu\nu} &= F^{tx}F_{tx} + F^{ty}F_{ty} + F^{tz}F_{tz} + F^{xt}F_{xt} + F^{xy}F_{xy} + F^{xz}F_{xz} \\ &\quad + F^{yt}F_{yt} + F^{yx}F_{yx} + F^{yz}F_{yz} + F^{zt}F_{zt} + F^{zx}F_{zx} + F^{zy}F_{zy} \\ &= -E_x^2 - E_y^2 - E_z^2 - E_x^2 + B_z^2 + B_y^2 - E_y^2 + B_z^2 + B_x^2 - E_z^2 + B_y^2 + B_x^2 \\ &= 2(B^2 - E^2) \end{aligned}$$

Where we used that $E^2 = E_x^2 + E_y^2 + E_z^2$ and $B^2 = B_x^2 + B_y^2 + B_z^2$.
Let us consider now a second-rank tensor defined as

$$T^{\mu\nu} = F^{\mu\alpha}F_{\alpha}^{\nu} + Cg^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}$$

We see that

$$F_{\alpha}^{\nu} = \eta_{\alpha\mu}F^{\mu\nu} = \begin{bmatrix} 0 & -E_x & -E_y & -E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{bmatrix}$$

Then if we compute T^{tt} in a LIF we get that

$$\begin{aligned} T^{tt} &= F^{t\alpha}F_{\alpha}^t + C\eta^{tt}F^{\alpha\beta}F_{\alpha\beta} \\ &= -E_x^2 - E_y^2 - E_z^2 - 2C(B^2 - E^2) \\ &= -E^2 - 2C(B^2 - E^2) \\ &= -2CB^2 + E^2(2C - 1) \end{aligned}$$

We need that $-2C = 2C - 1$ so $C = 1/4$ and hence must be that

$$T^{\mu\nu} = -\frac{1}{4\pi k} \left(F^{\mu\alpha}F_{\alpha}^{\nu} + \frac{1}{4}g^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta} \right)$$

So when $\mu = \nu = t$ we get that

$$T^{tt} = -\frac{1}{4\pi k} \left(-\frac{1}{2}(E^2 + B^2) \right) = \frac{1}{8\pi k}(E^2 + B^2) = \rho_E$$

as we wanted.

Also, we need to check if the definition of $T^{\mu\nu}$ we gave is symmetric, since the metric $g^{\mu\nu}$ is symmetric we only need to show that $F^{\mu\alpha}F_\alpha^\nu$ is symmetric.

We see that

$$F^{t\alpha}F_\alpha^t = -E_x^2 - E_y^2 - E_z^2 = -E^2$$

$$F^{t\alpha}F_\alpha^x = -E_yB_z + E_zB_y$$

$$F^{t\alpha}F_\alpha^y = E_xB_z - E_zB_x$$

$$F^{t\alpha}F_\alpha^z = -E_xB_y + E_yB_x$$

$$F^{x\alpha}F_\alpha^t = -B_zE_y + B_yE_z = F^{t\alpha}F_\alpha^x$$

$$F^{x\alpha}F_\alpha^x = E_x^2 - B_z^2 - B_y^2$$

$$F^{x\alpha}F_\alpha^y = E_xE_y + B_xB_y$$

$$F^{x\alpha}F_\alpha^z = E_xE_z + B_xB_z$$

$$F^{y\alpha}F_\alpha^t = E_xB_z - E_zB_x = F^{t\alpha}F_\alpha^y$$

$$F^{y\alpha}F_\alpha^x = E_yE_x + B_xB_z = F^{x\alpha}F_\alpha^y$$

$$F^{y\alpha}F_\alpha^y = E_y^2 - B_z^2 - B_x^2$$

$$F^{y\alpha}F_\alpha^z = E_yE_z + B_yB_z$$

$$F^{z\alpha}F_\alpha^t = -E_xB_y + E_yB_x = F^{t\alpha}F_\alpha^z$$

$$F^{z\alpha}F_\alpha^x = E_xE_z + B_xB_z = F^{x\alpha}F_\alpha^z$$

$$F^{z\alpha}F_\alpha^y = E_yE_z + B_yB_z = F^{y\alpha}F_\alpha^z$$

$$F^{z\alpha}F_\alpha^z = E_z^2 - B_y^2 - B_x^2$$

Therefore since $F^{\mu\alpha}F_\alpha^\nu$ is symmetric then $T^{\mu\nu}$ is symmetric.

b. Let us compute T^{tx} as follows

$$\begin{aligned} T^{tx} &= -\frac{1}{4\pi k} \left(F^{t\alpha}F_\alpha^x + \frac{1}{4}\eta^{tx}F^{\alpha\beta}F_{\alpha\beta} \right) \\ &= -\frac{1}{4\pi k} \left(E_zB_y - E_yB_z \right) \\ &= -\frac{1}{4\pi k} \left(\vec{E} \times \vec{B} \right)_x \end{aligned}$$

□

Solution. **P20.6**

- a. We know that if i and j are spatial coordinates then T^{ij} is the i -flux of j -momentum.

On the other hand, we saw that if a plate with area A is placed perpendicular to the x direction (in the path of the dust particles) then, for example, $T^{yx}A$ is the y -momentum per unit of coordinate time t .

Suppose we are in a LIF. If the plate is in any position we can generalize this by taking a unit normal four-vector perpendicular to A with components $\mathbf{n} = [0, n_x, n_y, n_z]$ where the time component is 0 since it's purely a spatial vector.

Then $T^{\mu\nu}An_\nu$ is the μ -momentum per unit of time, where μ is a spatial coordinate.

If μ is a time coordinate then $T^{t\nu}$ is the ν -flux of energy so $T^{t\nu}An_\nu$ is the ν -energy per unit of time. Therefore the four-force $\mathbf{F}$ is given by

$$\mathbf{F} = \left(\frac{d\mathbf{p}}{d\tau} \right)^\mu = T^{\mu\nu} An_\nu = T^{\mu\nu} A \eta_{\nu\alpha} n^\alpha$$

Where we added in the last step $\eta_{\nu\alpha}$ since we are working on a LIF and it doesn't change the equation since the time component of $\mathbf{n}$ is 0.

Finally, we can generalize this, using the appropriate metric as

$$\mathbf{F} = \left(\frac{d\mathbf{p}}{d\tau} \right)^\mu = T^{\mu\nu} A g_{\nu\alpha} n^\alpha$$

- b. Let us consider a perfect fluid. If $\mathbf{n}$ represents a wall in space and $\mathbf{u}$ represents the fluid's four velocity then if the fluid is at rest with respect to the wall we must have that $\mathbf{u} \cdot \mathbf{n} = g_{\nu\alpha} u^\nu n^\alpha = 0$ since the only non-zero component of u^ν is $u^t = 1$ but $n^t = 0$.

Then the four-force exerted on any patch of area A on the wall of the container holding the perfect fluid at rest is

$$\begin{aligned} \mathbf{F} &= \left(\frac{d\mathbf{p}}{d\tau} \right)^\mu = T^{\mu\nu} A g_{\nu\alpha} n^\alpha \\ &= [(\rho_0 + p_0)u^\mu u^\nu + p_0 g^{\mu\nu}] A g_{\nu\alpha} n^\alpha \\ &= 0 + p_0 g^{\mu\nu} A g_{\nu\alpha} n^\alpha \\ &= p_0 A \delta^\mu_\alpha n^\alpha \\ &= p_0 A n^\mu \end{aligned}$$

Therefore the magnitude of the four-force exerted on any patch of area A on the wall of the container is $p_0 A$.

□

Solution. **P20.7**

Let us consider the tensor $T^{\mu\nu} = -\Lambda g^{\mu\nu}$ if we take a LIF then the tensor becomes $T^{\mu\nu} = -\Lambda \eta^{\mu\nu}$ i.e.

$$T^{\mu\nu} = \begin{bmatrix} \Lambda & 0 & 0 & 0 \\ 0 & -\Lambda & 0 & 0 \\ 0 & 0 & -\Lambda & 0 \\ 0 & 0 & 0 & -\Lambda \end{bmatrix}$$

Comparing this Stress-Energy tensor to the one we found for a perfect fluid then we see that the pressure $p_0 = -\Lambda$ and since Λ is a positive scalar then the pressure is negative, so if $T^{\mu\nu}$ is a Stress-Energy tensor of a real fluid must be, for example, that

$$T^{xx} = -\Lambda = \frac{1}{L^3} \iiint N(|\vec{p}|) \frac{(p^x)^2}{p^t} dp^x dp^y dp^z$$

But $N(|\vec{p}|) \geq 0$, $(p^x)^2 \geq 0$ and $p^t = m/\sqrt{1-v^2}$ which is also positive, so the integral must be positive as well.

Therefore $T^{\mu\nu} = -\Lambda g^{\mu\nu}$ cannot represent a real fluid's Stress-Energy tensor. $\square$

Solution. **P20.8**

- a. Let us consider the second-rank tensor $M_{\alpha\beta,obs}$ defined as the tensor product of two four-vectors

$$\begin{aligned} M_{\alpha\beta,obs} &= A_{\alpha,obs} B_{\beta,obs} \\ &= (\mathbf{o}_\alpha)^\sigma g_{\sigma\gamma} A^\gamma (\mathbf{o}_\beta)^\lambda g_{\lambda\delta} B^\delta \end{aligned}$$

We also note that

$$\begin{aligned} M_{obs}^{\mu\nu} &= \eta^{\mu\alpha} \eta^{\nu\beta} M_{\alpha\beta,obs} \\ &= \eta^{\mu\alpha} \eta^{\nu\beta} (\mathbf{o}_\alpha)^\sigma g_{\sigma\gamma} A^\gamma (\mathbf{o}_\beta)^\lambda g_{\lambda\delta} B^\delta \\ &= \eta^{\mu\alpha} \eta^{\nu\beta} A^\gamma B^\delta g_{\gamma\sigma} (\mathbf{o}_\alpha)^\sigma g_{\delta\lambda} (\mathbf{o}_\beta)^\lambda \\ &= \eta^{\mu\alpha} \eta^{\nu\beta} M^{\gamma\delta} g_{\gamma\sigma} (\mathbf{o}_\alpha)^\sigma g_{\delta\lambda} (\mathbf{o}_\beta)^\lambda \end{aligned}$$

Therefore for the stress-energy tensor we get that

$$T_{obs}^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} T^{\gamma\delta} g_{\gamma\sigma} (\mathbf{o}_\alpha)^\sigma g_{\delta\lambda} (\mathbf{o}_\beta)^\lambda$$

- b. Let us consider an observer with four-velocity $\mathbf{u}_{obs}$ with respect to the local rest frame of a perfect fluid.

We know that the stress-energy tensor of the perfect fluid is given by equation (20.15) so in the observer's frame the energy density is

$$\begin{aligned} T_{obs}^{tt} &= \eta^{tt} \eta^{tt} T^{\gamma\delta} g_{\gamma\sigma} (\mathbf{o}_t)^\sigma g_{\delta\lambda} (\mathbf{o}_t)^\lambda \\ &= T^{\gamma\delta} g_{\gamma\sigma} (\mathbf{u}_{obs})^\sigma g_{\delta\lambda} (\mathbf{u}_{obs})^\lambda \end{aligned}$$

Where we used that $\mathbf{o}_t = \mathbf{u}_{obs}$, also, we see that in the rest frame of the perfect fluid $\mathbf{u}_{obs}$ is

$$(\mathbf{u}_{obs})^\mu = \begin{bmatrix} \gamma \\ \gamma v_x \\ \gamma v_y \\ \gamma v_z \end{bmatrix}$$

Also, since the local rest frame is inertial $g_{\mu\nu} = \eta_{\mu\nu}$, hence

$$\begin{aligned} T_{obs}^{tt} &= T^{\gamma\delta} \eta_{\gamma\sigma} (\mathbf{u}_{obs})^\sigma \eta_{\delta\lambda} (\mathbf{u}_{obs})^\lambda \\ &= T^{tt} \gamma^2 + T^{xx} \gamma^2 v_x^2 + T^{yy} \gamma^2 v_y^2 + T^{zz} \gamma^2 v_z^2 \\ &= \rho_0 \gamma^2 + p_0 \gamma^2 v^2 \\ &= \rho_0 \gamma^2 + p_0 \gamma^2 v^2 + \gamma^2 p_0 - \gamma^2 p_0 \\ &= \rho_0 \gamma^2 - p_0 \gamma^2 (1 - v^2) + \gamma^2 p_0 \\ &= (\rho_0 + p_0) \gamma^2 - p_0 \end{aligned}$$

Finally, if we assume the fluid in the observer's frame moves at the same speed but in the opposite spatial direction, then the component u^t of the fluid velocity in the observer's frame is $u^t = \gamma$ and from equation (20.16) we get that

$$T^{tt} = (\rho_0 + p_0) (u^t)^2 + p_0 \eta^{tt} = (\rho_0 + p_0) \gamma^2 - p_0$$

□

Solution. P20.9

We know that

$$\rho_{obs} = T^{\mu\nu} \eta_{\mu\alpha} \eta_{\nu\beta} u_{obs}^{\alpha} u_{obs}^{\beta}$$

and that

$$T^{\mu\nu} = \begin{bmatrix} \rho & 0 & 0 & 0 \\ 0 & \alpha\rho & 0 & 0 \\ 0 & 0 & \alpha\rho & 0 \\ 0 & 0 & 0 & \alpha\rho \end{bmatrix} \quad \mathbf{u}_{obs}^{\alpha} = \gamma [1, v_x, v_y, v_z]$$

Then in the original LIF

$$\begin{aligned} \rho_{obs} &= T^{tt}(u_{obs}^t)^2 + T^{xx}(u_{obs}^x)^2 + T^{yy}(u_{obs}^y)^2 + T^{zz}(u_{obs}^z)^2 \\ &= \rho\gamma^2 + \alpha\rho\gamma^2 v_x^2 + \alpha\rho\gamma^2 v_y^2 + \alpha\rho\gamma^2 v_z^2 \\ &= \rho\gamma^2 + \alpha\rho\gamma^2 v^2 \\ &= \rho\gamma^2 + \alpha\rho\gamma^2 v^2 + \alpha\rho\gamma^2 - \alpha\rho\gamma^2 \\ &= \rho\gamma^2 - \alpha\rho\gamma^2(1 - v^2) + \alpha\rho\gamma^2 \\ &= \rho\gamma^2(1 + \alpha) - \alpha\rho \end{aligned}$$

We want that $\rho(\gamma^2(1 + \alpha) - \alpha) > 0$ then assuming $\rho > 0$ must be that

$$\begin{aligned} \gamma^2\alpha - \alpha &> -\gamma^2 \\ \alpha(\gamma^2 - 1) &> -\gamma^2 \\ \alpha &> -\frac{\gamma^2}{\gamma^2 - 1} \\ \alpha &> \frac{1}{(1 - v^2)(1 - \frac{1}{1-v^2})} \\ \alpha &> \frac{1}{(1 - v^2) - 1} \\ \alpha &> -\frac{1}{v^2} \end{aligned}$$

Since $\alpha > -1/v^2$ needs to hold for every speed if we set $v = 1$ we get that

$$\alpha > -1$$

□